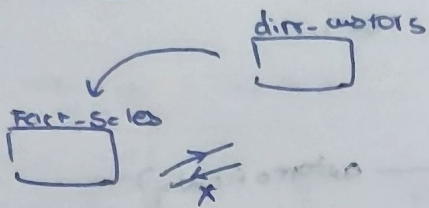


12 Sept

①

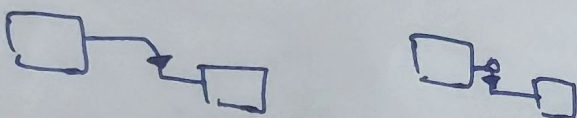
- in attributing filtering

Best Practices

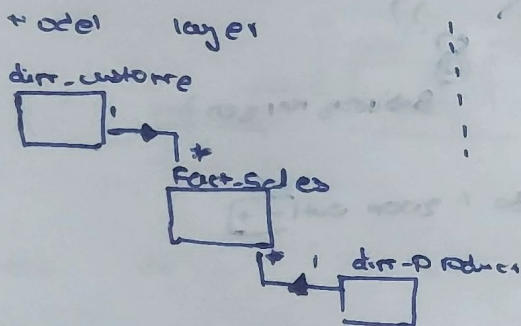


① Design Model view $\Rightarrow$ Verify in Report view

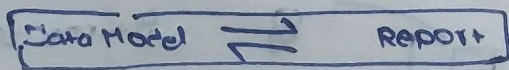
- Good control of behaviour Data Model (advice patterns)
- Don't make dimension to fact table



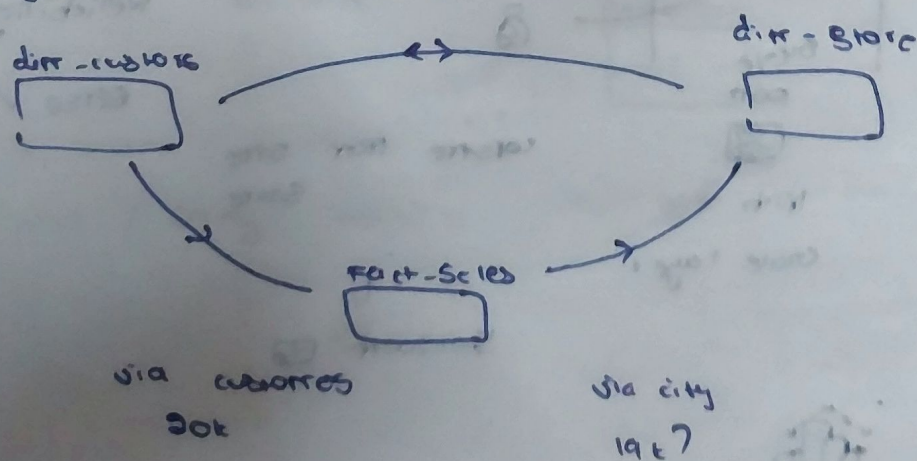
Eg.: Star schema xx ——— xx ——— xx ^{Unusual cases!!} Revisior-1
 Model layer Report layer $\rightarrow$ 200000 150.00



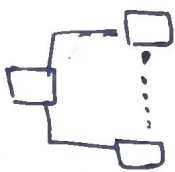
- control the behaviour of Data Model
- good report



Types of Relationship

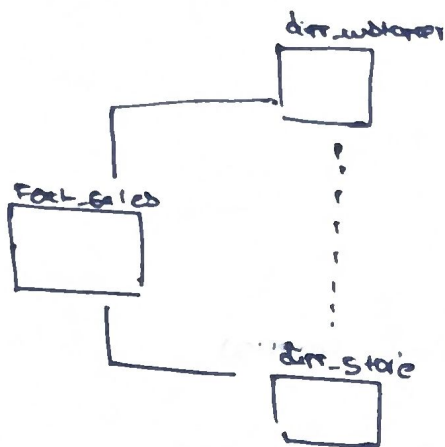


pbi doesn't know the part two tables

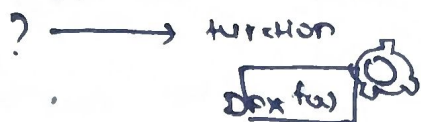
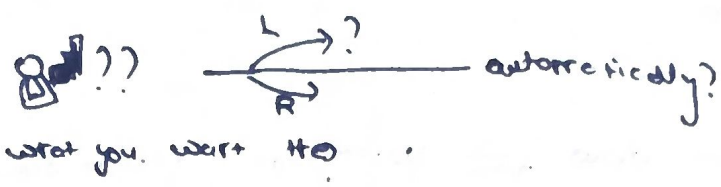


Revision - 1

Relationship



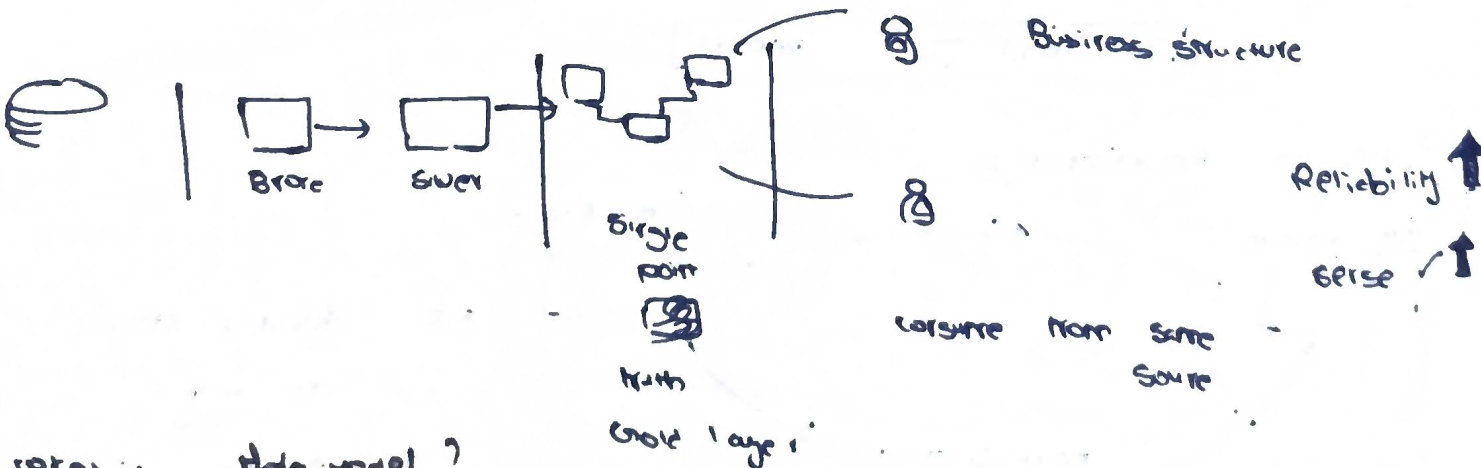
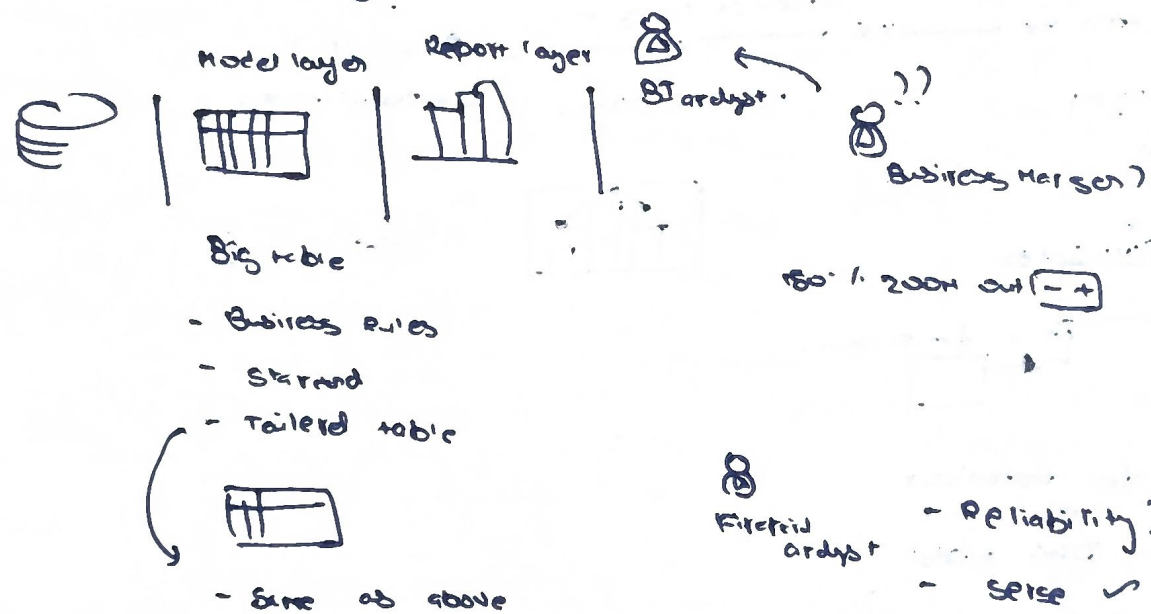
... + 200m in 150'1.



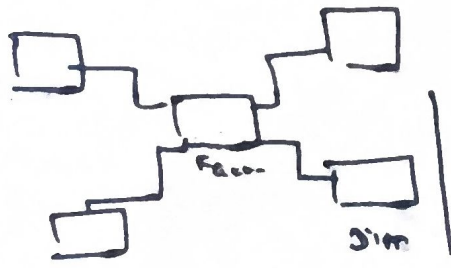
Revision-3 !!

why is

Data Model?



what is data model?



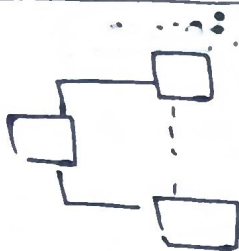
150'1.200m

Business data & how the pieces are related

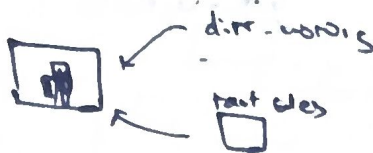
Report layer



Type ? why is Needed



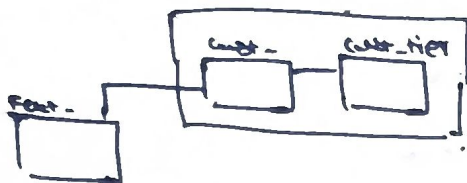
BI? Day out?



- automatic
- force

xx when 1 to 1 is harmless

Edge case !!



- merge
- split

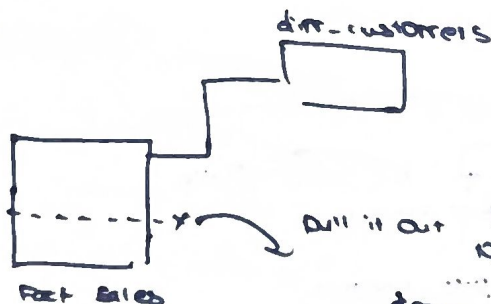
Sensitive
Big Dimension

access control (Row level Security?)

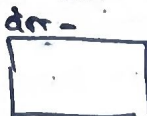
Data Minimization

xx 10/9/06

Turn dimension



Put it out to create new



Id + dates + numbers

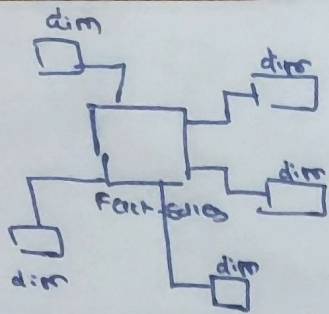
text / Entry usually

describe something

(One candidate for dimension)

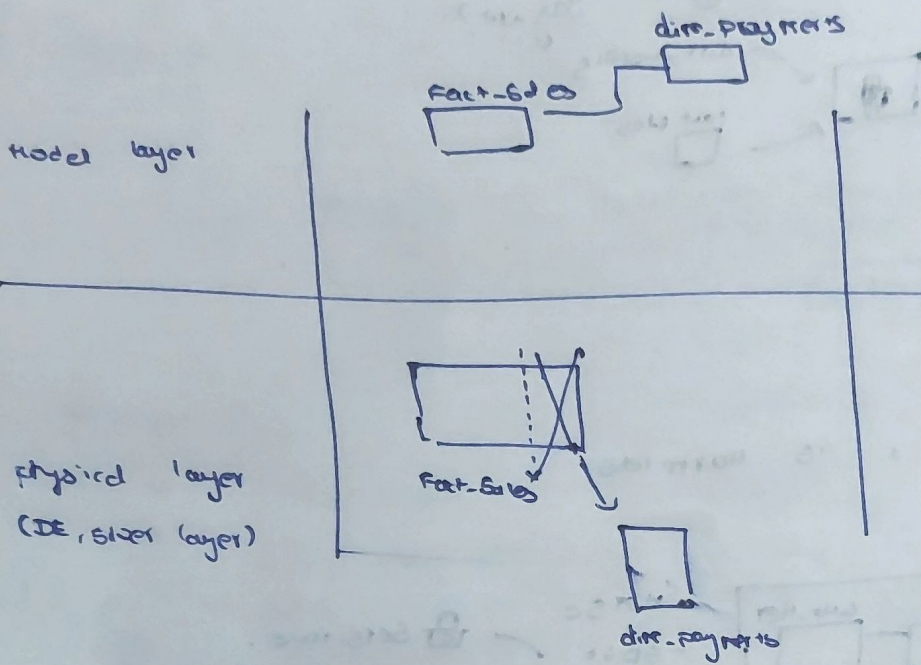
Reliability ↑
Merge ✓ ↑

has
pieces
mark



Turn dimension

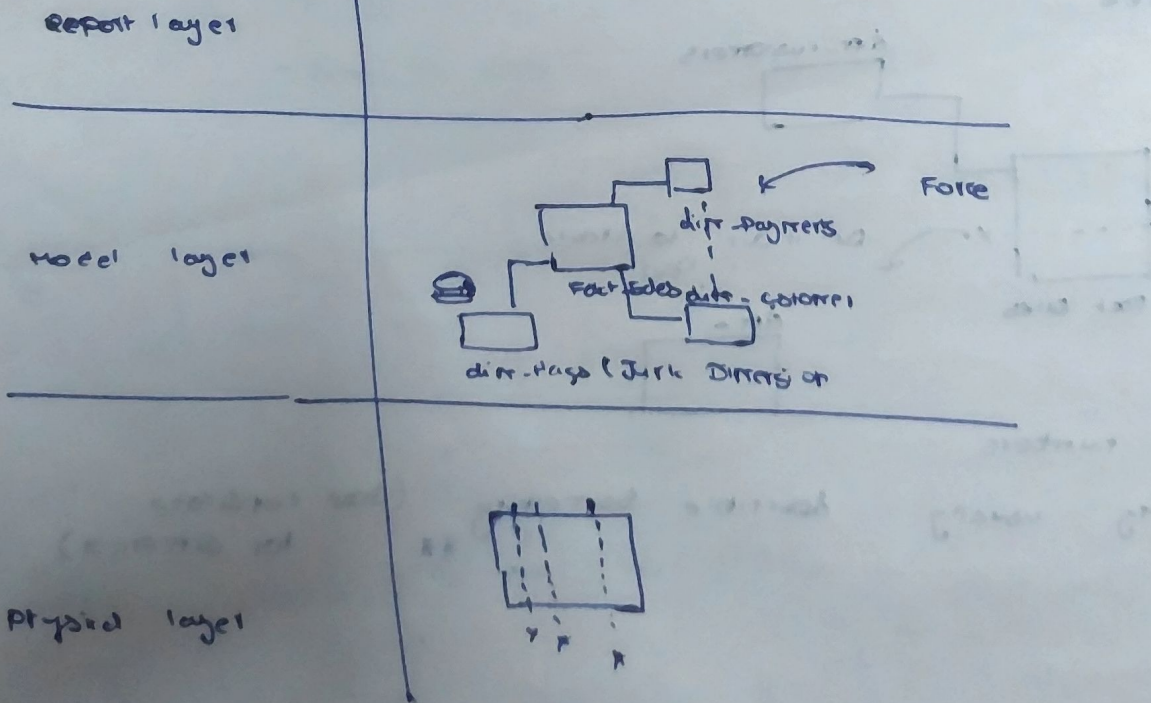
group unrelated dimension



Revision-1

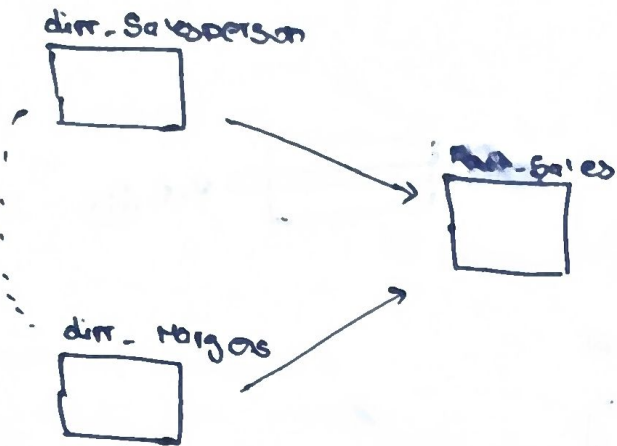
look up $\rightarrow$ extract
Surrogate
key

Revision-2



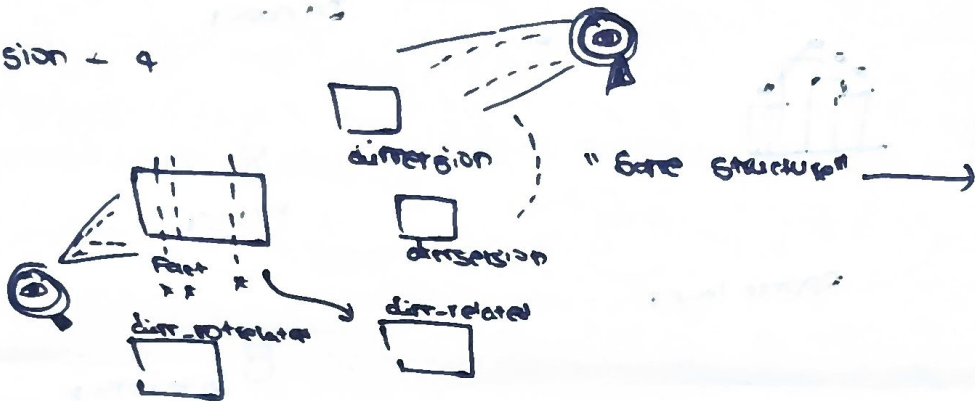
- Investigate the data itself

Task: : Calculate sales by managers



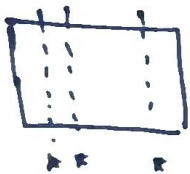
'Same structure'
Same schema

Revision - 4

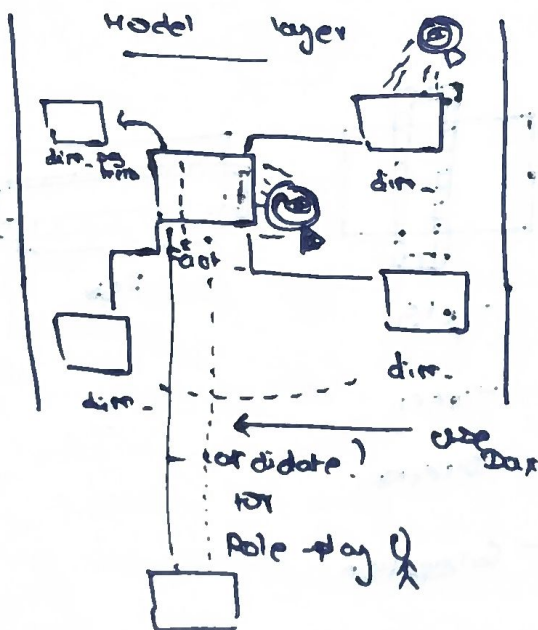


Revision - 3

Physical layer



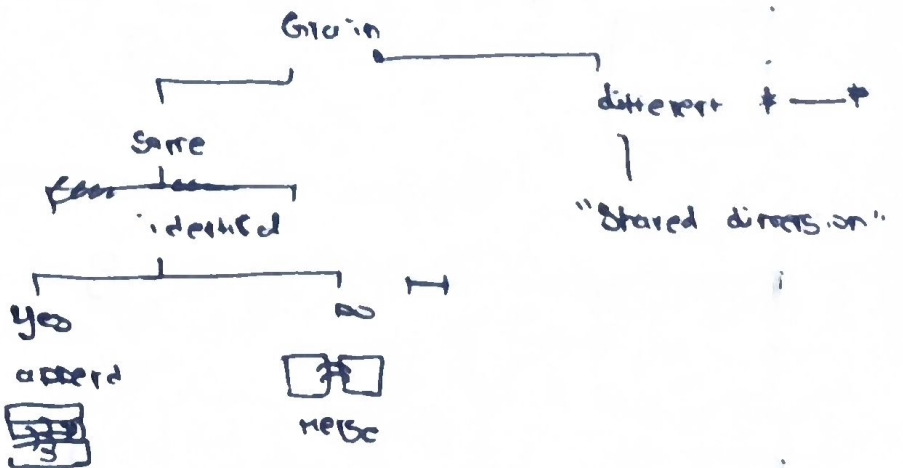
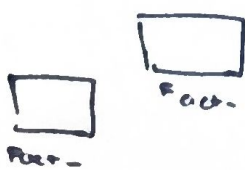
Model layer



Report layer



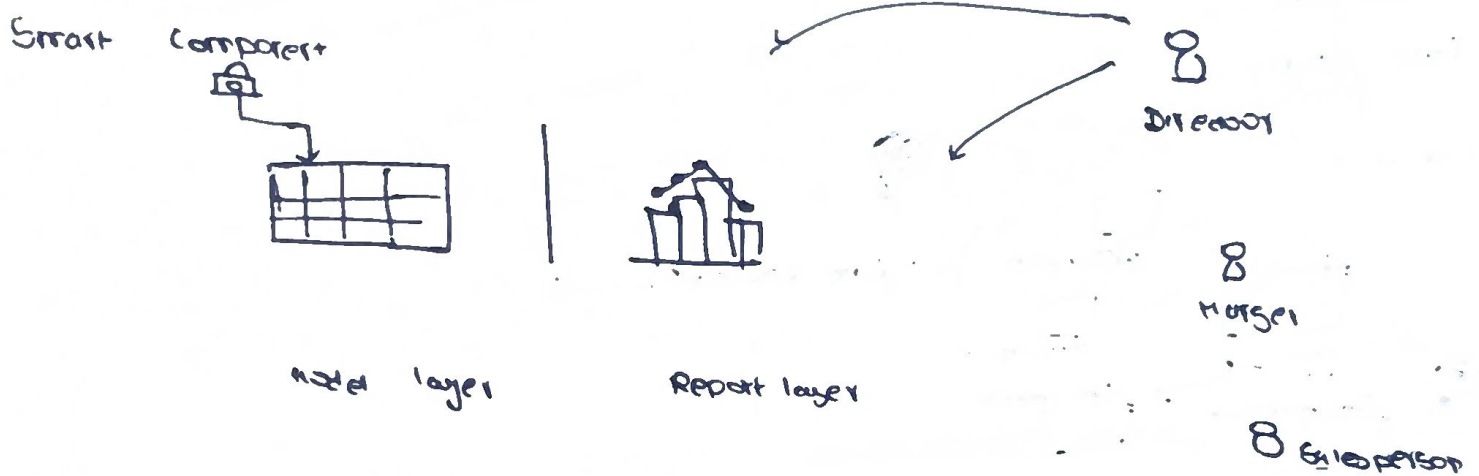
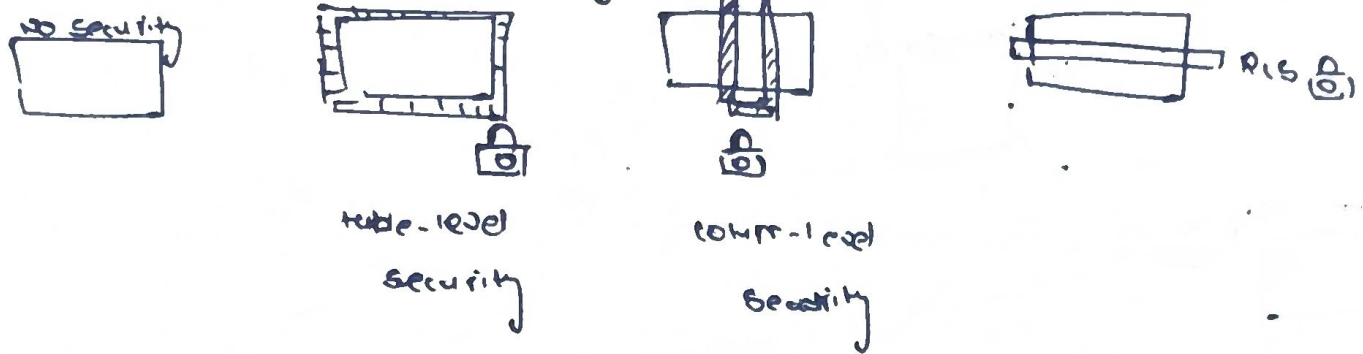
12/9/20



Security in Data Modeling

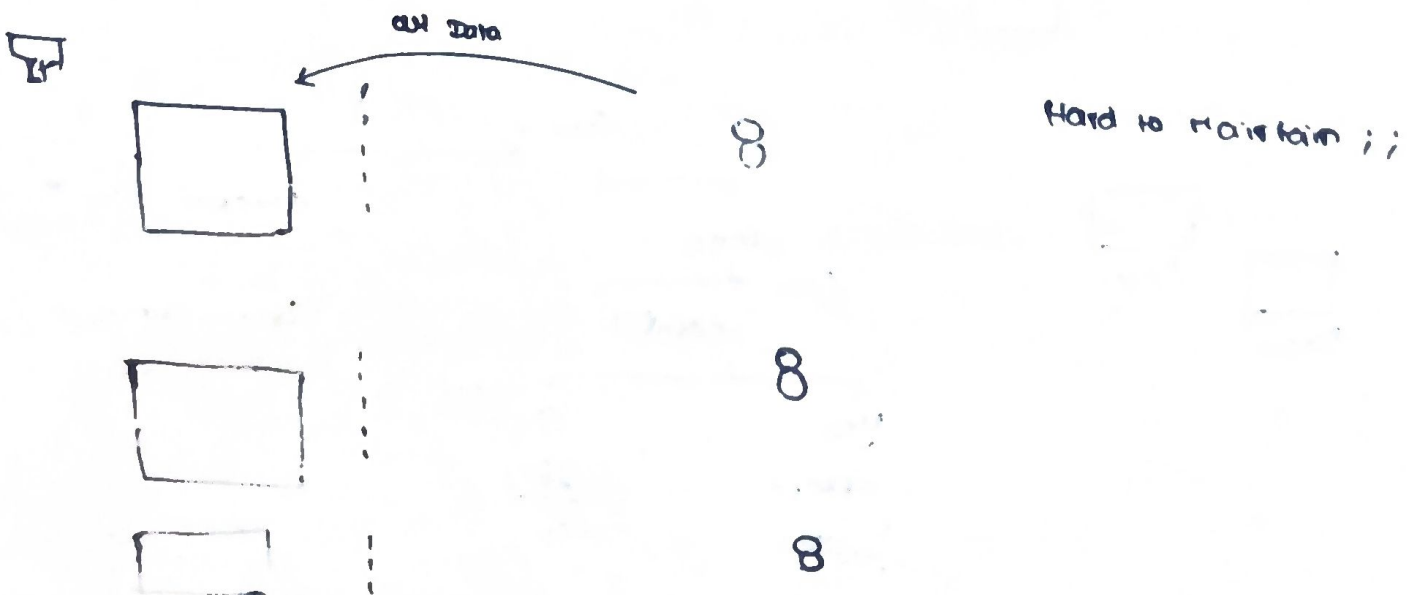
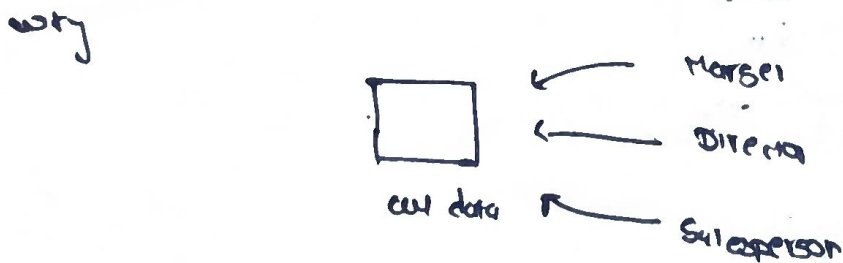
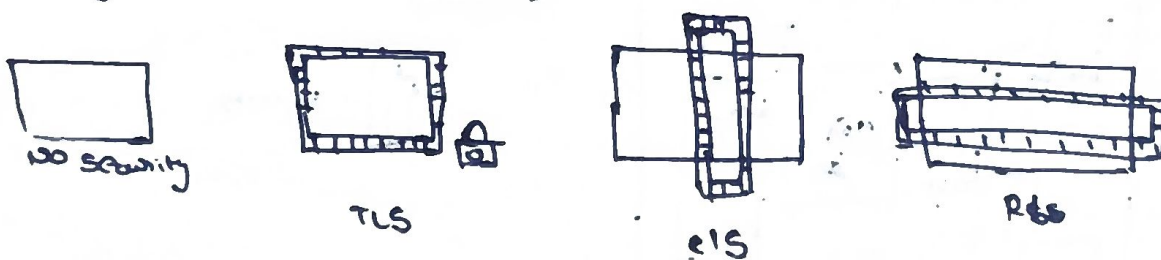
clear model $\longrightarrow$ easy to implement security

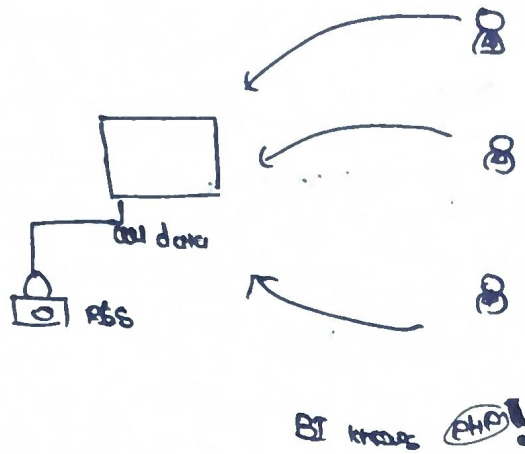
fuzzy model $\longrightarrow$ hard to implement security



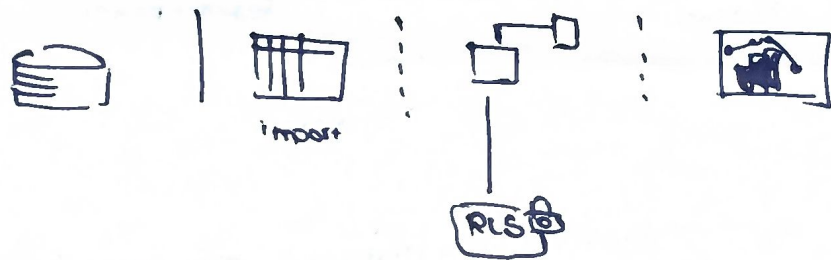
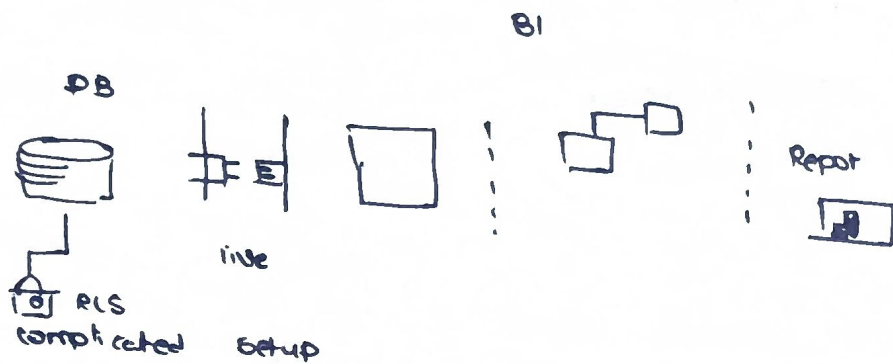
Revision - 1

Security & its types

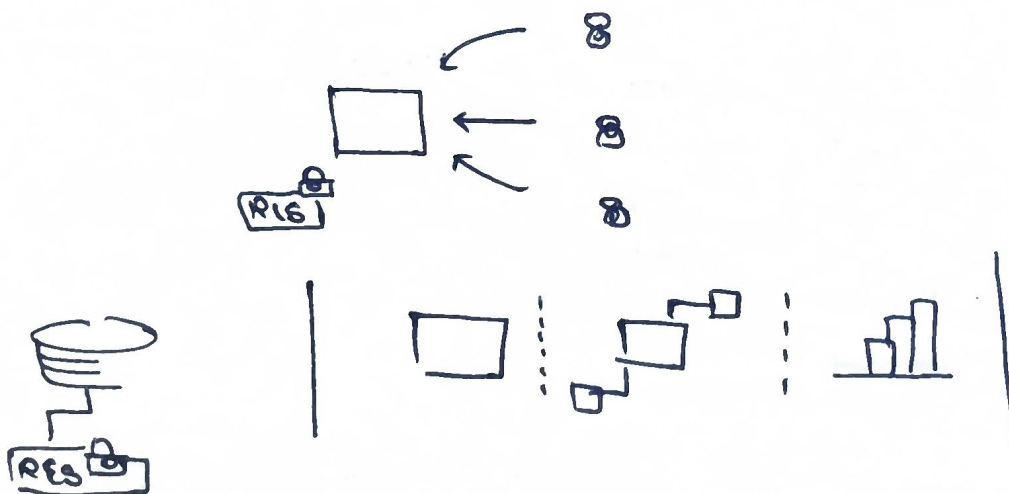




— how to implement it



— Revision 3



we

1000

Marked step

Red-Box → "EV"

Rep. $\rightarrow$ US!

maria

REVISION- 9 - END - 14/9/56

Marger

61 expression

Director

۱۵۵

- नाइफ



← director

8



← salesperson

"Hard to Maintain"

P. Service

Prüfung: $\text{Prüfung} = \{ \text{all-}, \text{us-} \}$



Rules - 1 = 'عد'

$$R_{\alpha\beta} - \gamma = 'Eu'$$

Marsen

10

as Bet

Marger

dirty security

১৫

1419150

①

Grain?

- level of detail - ore, row in fact represent

Fact - to - Fact

Grain - ✓

Area - ✓

Schema - ✓

One to One

added
+
Source

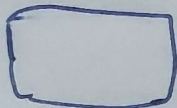
fact - Sales

fact - Sales -
US

fact - Sales

fact - Sales

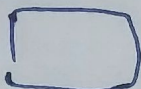
fact - combined



fact - Sales

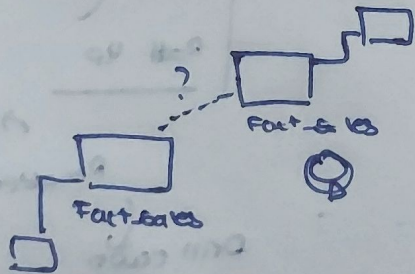


dim -

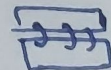


fact -

Revision -



Grain Dimension 11



level 1

level 2

level 3

Roll up

total

measure

recorded

category
record

Drill down

fact

product

Revisior-3

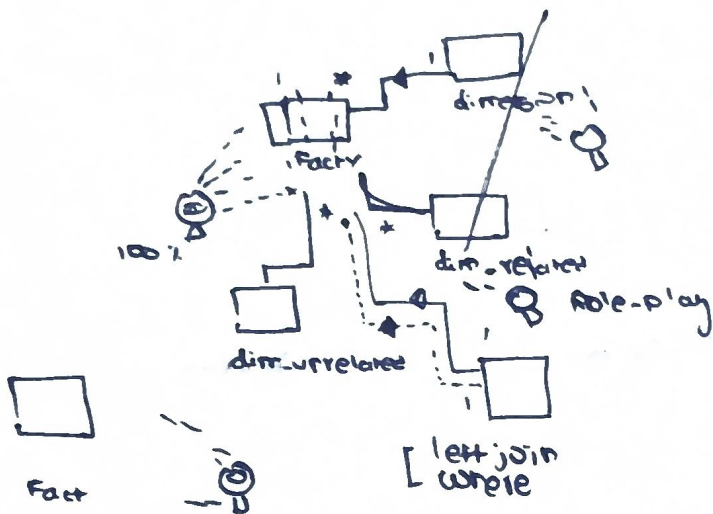
Physical



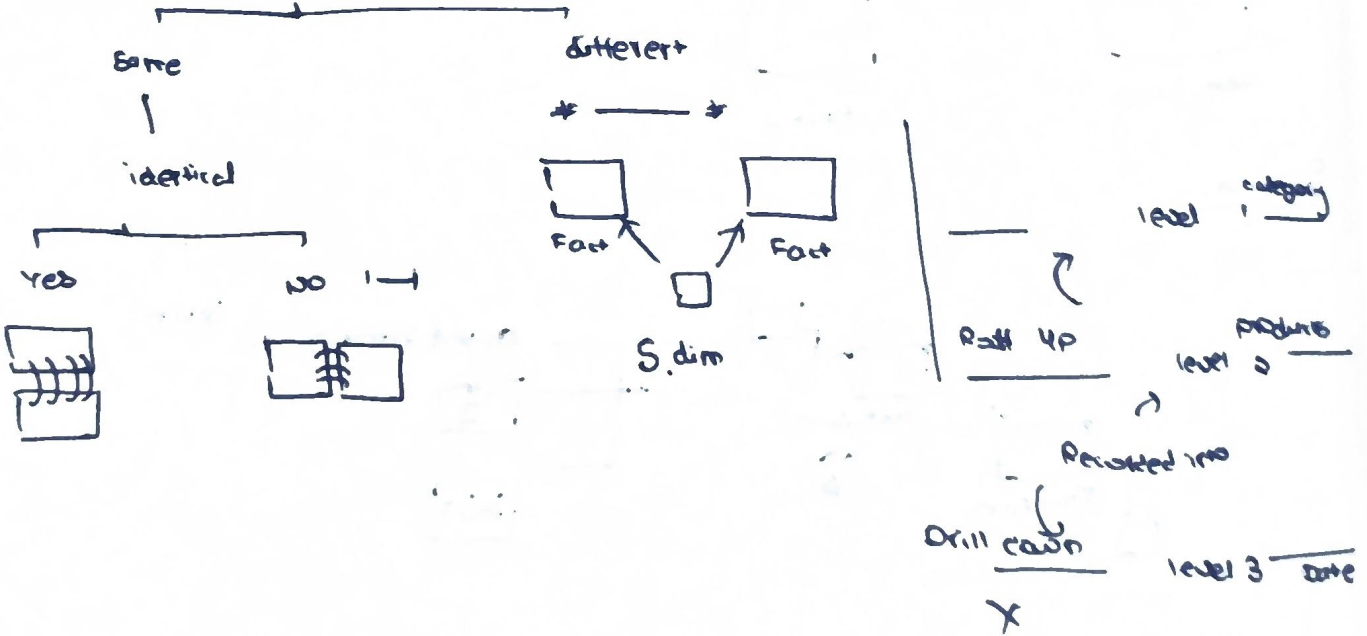
Model

Report





Grain - level 2 into of each row



15/11/20

①

New use

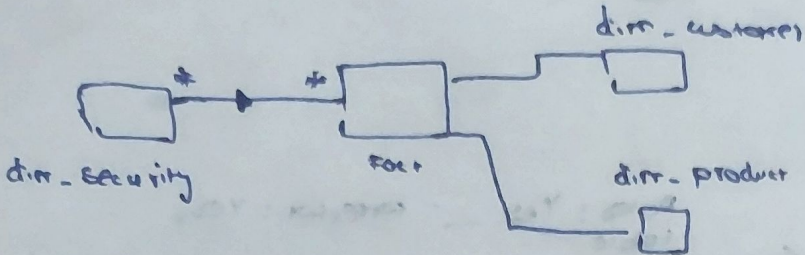
→ assign

→ Scalability

Security table:

S. connector:

idea as:



understand security

requirement or the right

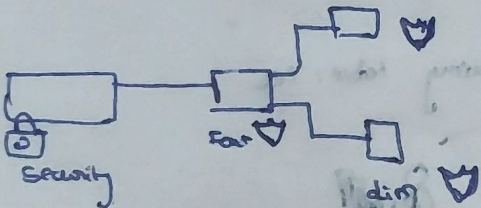
??

Security officer

keep in your mind 11, 85 number

10min

Revisit - 1

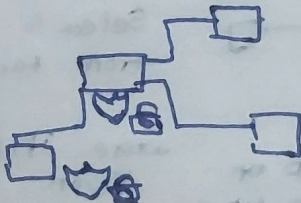


Security is part of data model

①

understand security req. of project

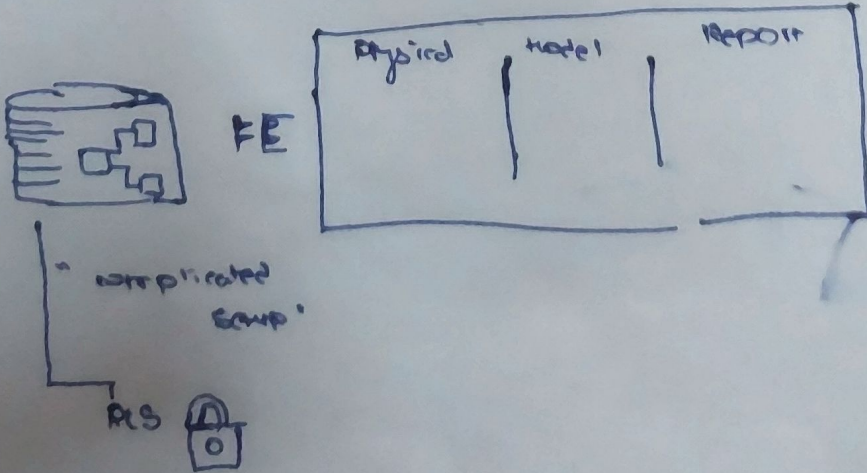
Security officer

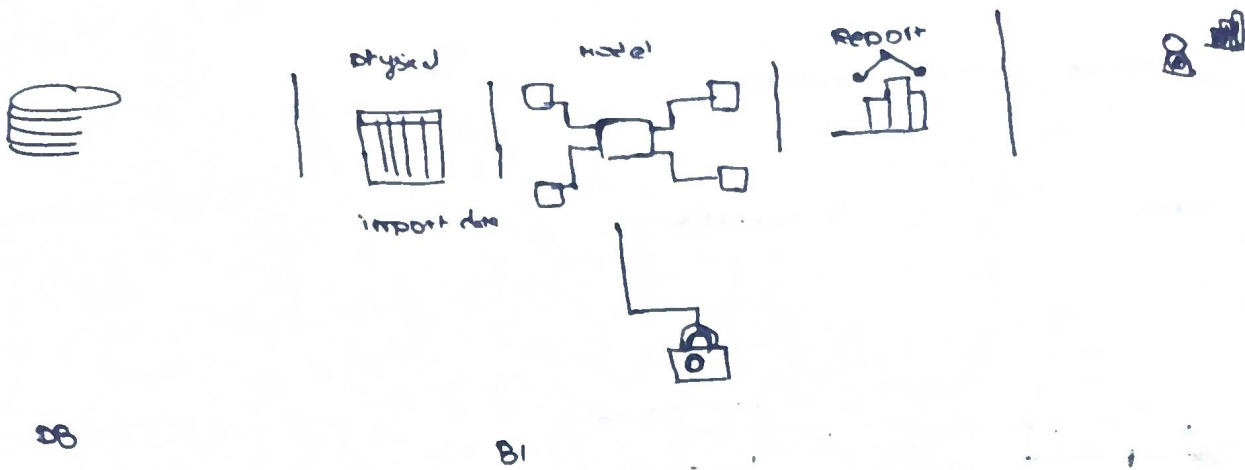


→ A/B testing
→ Big number

same only what's needed

How?

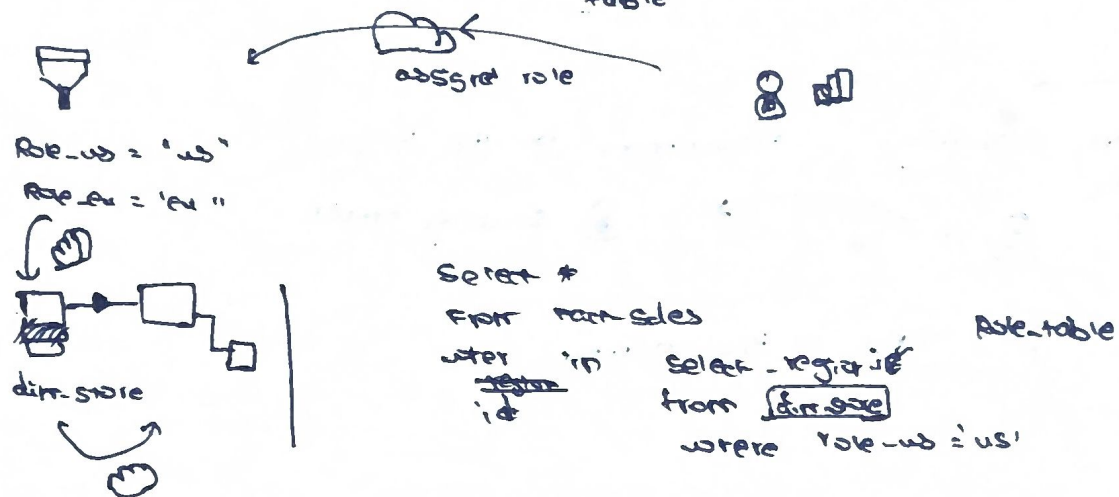




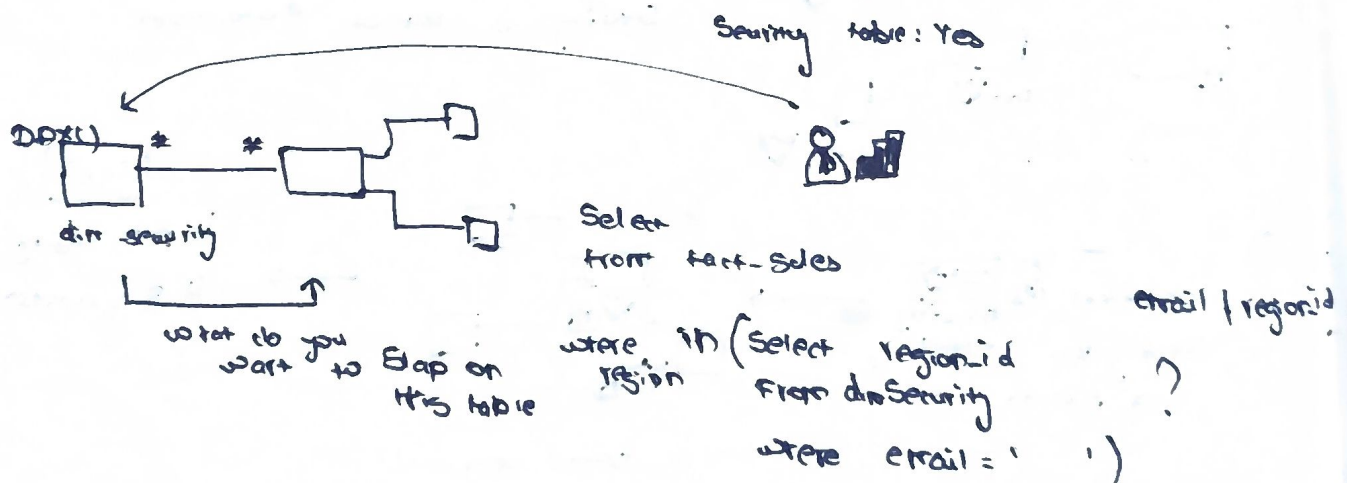
Static - RLS

Roles: Yes
table

correctness: Yes



Dynamic RLS



16 Sep + 20

①

We Fine Data Modeling

- ① Prepare & understand the data
- ② Spot dimension & Fact

③ Dimensions

④ Fact & connect

⑤ polish the data model

Rules

① Build S-Extra

② always understand the grain

③ every column must earn its space

④ Protect your numbers

Naming convention

① language + Statecode

② prefix & suffix

③ -id - name source

-key - we make it

④ prize

||

👉 Price

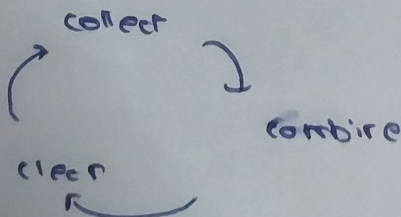
||

<> goods

— ④ →

20 min

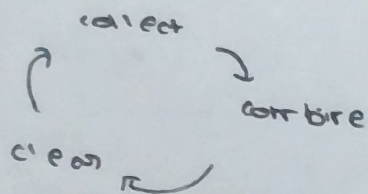
Dimension



40 min

17/9/26

①



CUSTOMER MASTER

cities

customer - contact

user - details

region

addresses

dim - customers

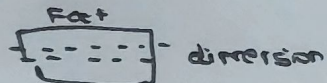
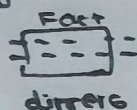
- Standardization

① orderly by related info

② verapre ABC

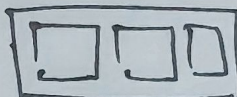
Building Facts

way 1



way 2

collect all facts together

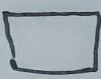


Event (Fact)



Fact

dim



dim

dim

header

Details

Fact - orders

Fact - order - line



Star report



whole number

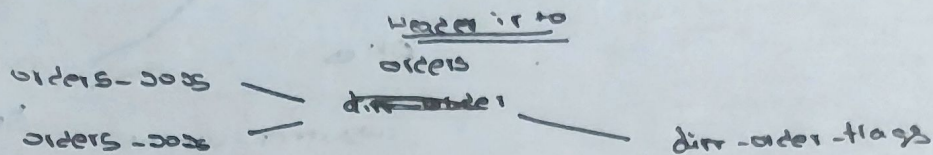


corrects reports

we Filter Data like crozy

fact table always sits at the level of measure

where do the number comes from



① Rules :: protect your numbers

- row total
- protect other charge

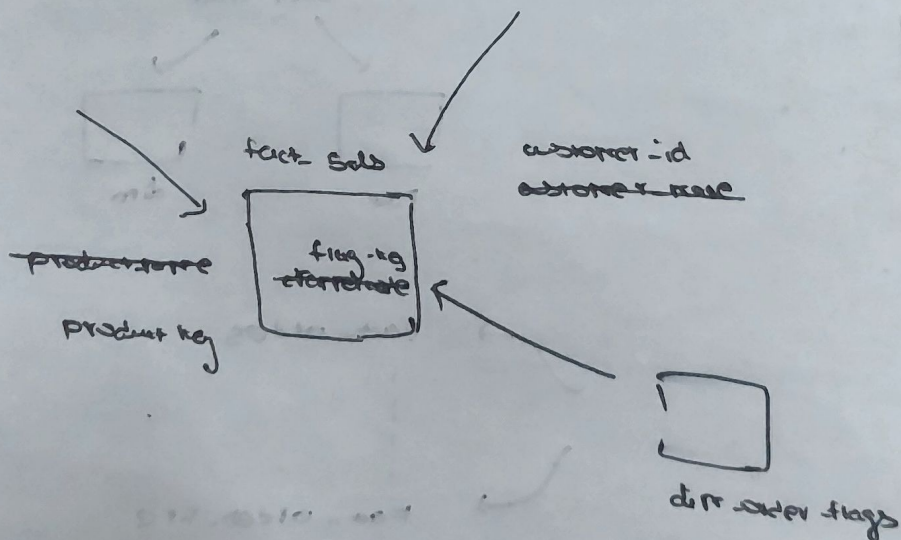
1:17
Project - silver layer

Grain (one row represents)

↳ Hospital & its spec

dim-product

dim-customer



18/9/20

when correlating with "Jure Dimension"

It will behave differently from Source-ids



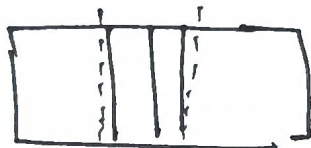
are count, are place
- Reduce the noise from source system

Eg customer city in data-customer



Every count must fall to place
+ variance ABC
+ records 100

fact-sales



ship to city
bill to city

data go



1:28:15,,

- Restore Data

by changing the layout of the data without losing
the meaning and the value of your data inside table

- If we have multiple rows in single column

column
Jan Sales

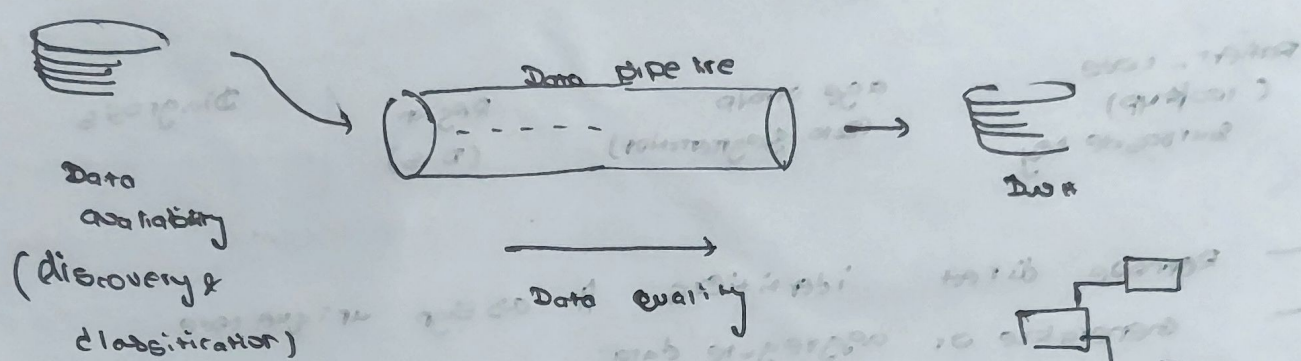
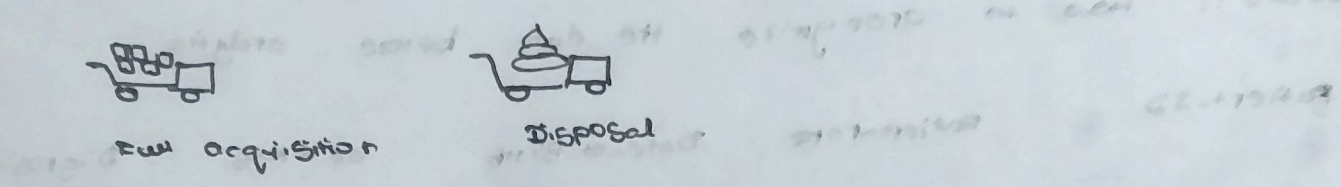
Split

column 1	column 2
Jan	Sales

Data Governance:

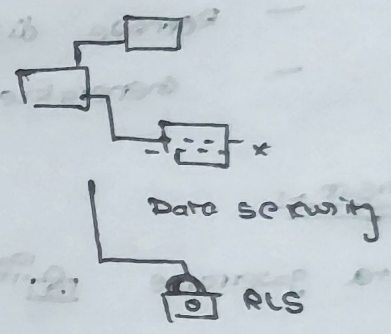
Flow Rules/ process to product our data

Data life cycle

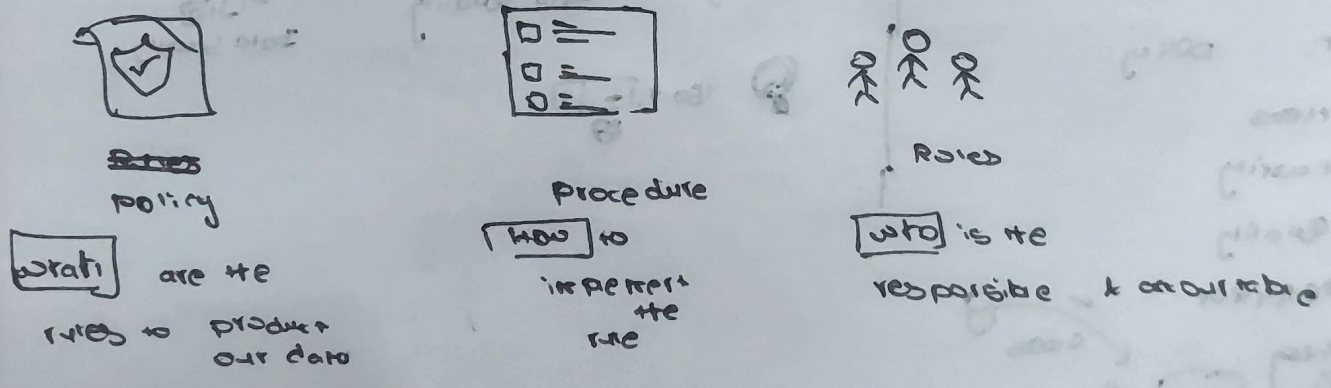


Data compliance

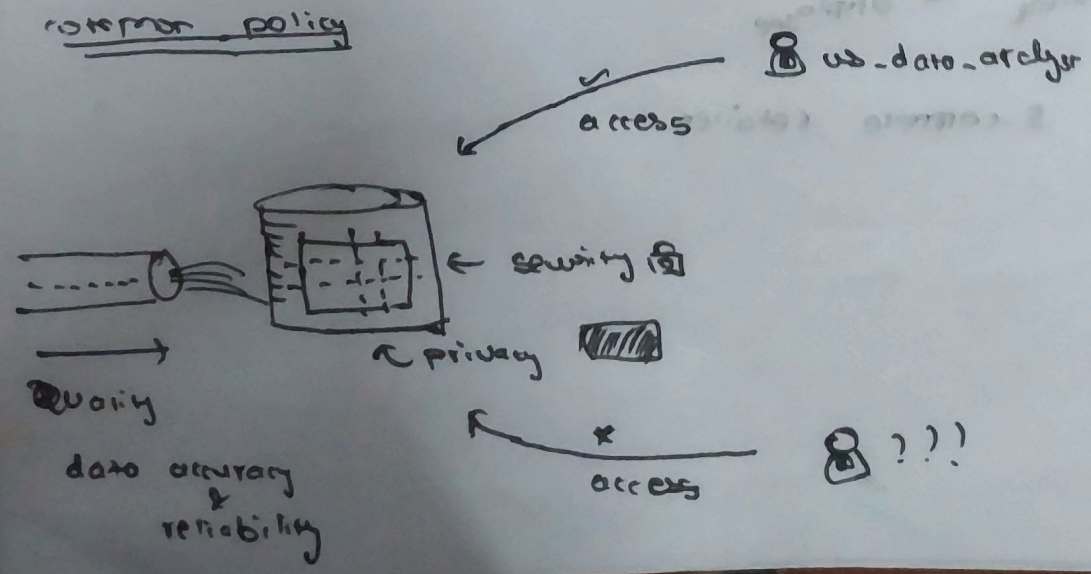
- ☒ Regulatory requirements
- ☒ Safety industry std
- ☐ Security



Key components



Eg: Make watermark into common policy



Roles → Data Owner
→ Data Steward

Eg: Hospital

Policy :: HIPAA → no law for protecting patient info

Procedure :: how to anonymize the data before analyzing

Patient-ID Patient-name Date-of Birth address Diagnosis

↓

Patient-code, (lookup) surrogate key age group (etc. generalization) Region (D.S.) Diagnosis

- Remove direct identifiers // assign unique code
- generalize or aggregate data

Revisiting

• Data governance

• Key components

Policy (what?)

Procedure (how?)

Roles (who?)

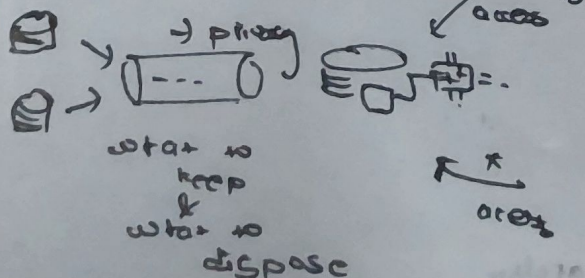
• Common Policy

→ access

→ security

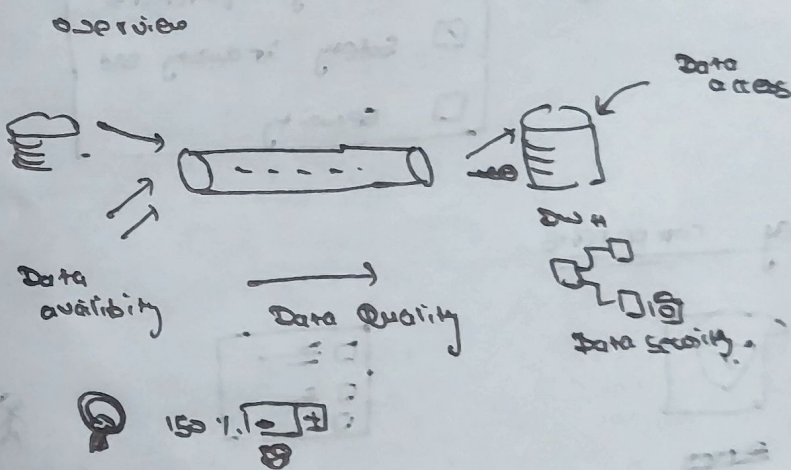
→ Quality

→ Retention



Eg: Hospital

& E-commerce retailer



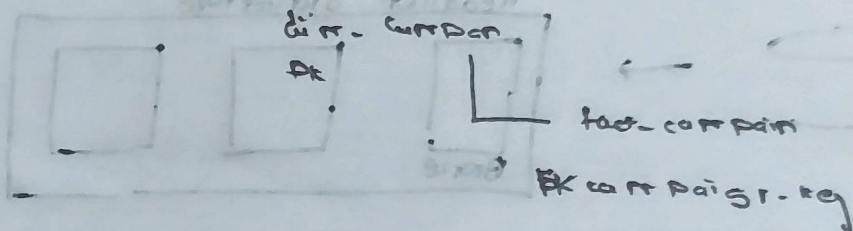
19 Sept 26

①

dimension is context // Data type
 fact is event }
 one describe, one group

inventory → fact-inventory ← dim-products

campaign-log — dim-campaign
 — fact-campaign-spent

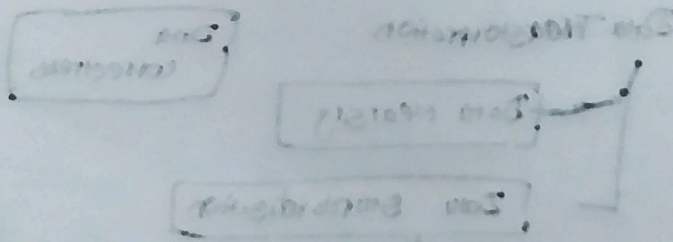


→ factless fact (fact without measure)

campaign - Spent → fact-promotion - converge

— Data Quality
 Can we trust our data?

Id Name date amount itype location advertiser



- completeness
- Validity
- uniqueness
- consistency
- time trends

Q.M. technique

- Data profiling (evaluate its context & structure)

- Data cleaning { Removing duplicates
 data consistency
 irrelevant observation

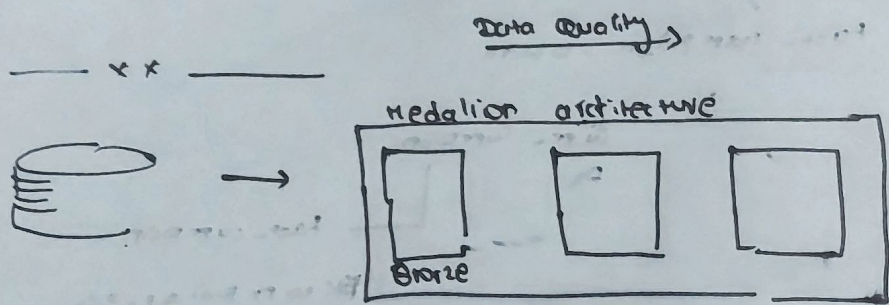
- Data Normalization { derive column
 splits column

Data validation — accuracy checks
 at various first errors
 → negative amount

Data consistency — Referential integrity checks

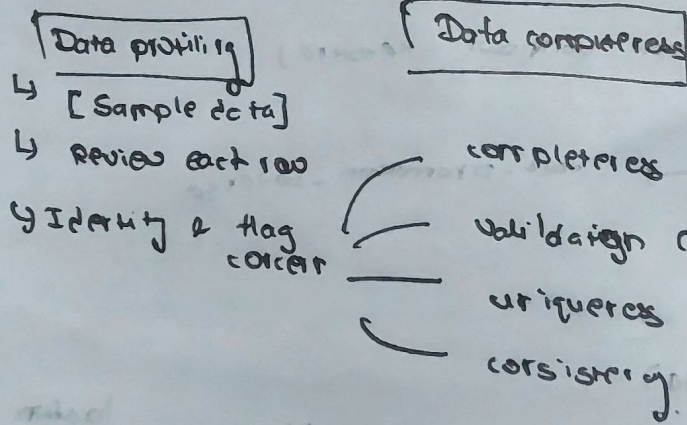
Master data Management ::

ensuring are consist is of "master data" SCD?



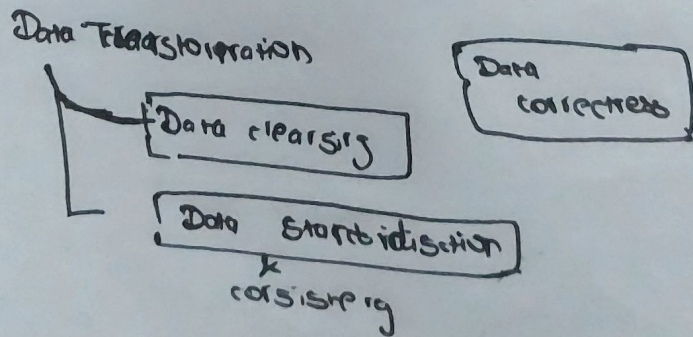
analyse → coding → validating → Documenting

- interview
source
expert

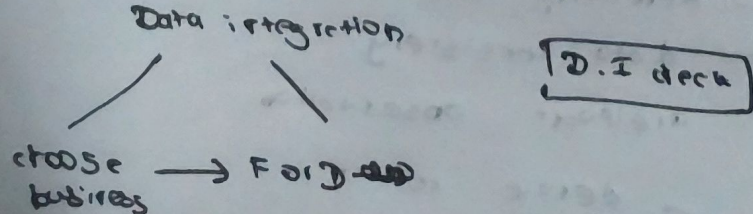


analyse → coding → validating → Documenting

Expert
understand
the business
data



analyse → coding → validating → Documenting



Continuous Monitoring

Data profiling ::

Data completeness ::

Data transformation ::

Data cleansing ::

Data validating ::

Data collection ::

User feedback loop ::

Table

Metric

— x x —

Soft Skills

01) D. Platform architecture

02) Data governance, catalog & contracts

03) Design Data model & Schema

Documenting

B.

— x x Accumulating Snapshot Fact

→ one row per event

→ contains all milestone date at every step

key

milestone date

Measure

Start with Spire

Orders

Payments

Invoices

Shipments

fact_order_payments

S

Documenting

we

protect the core

↳ understand the ~~best~~ business

↳ build only fact & dimensions

↳ Separate data model for special cases

↳ protect the core

PRE
... SE

xx ——— PROPOSE 11:::

0:07

